

Development Blueprint for NearDrop: A Modern Cross-Platform File-Sharing Ecosystem

Project Vision and Objective

The primary objective of this project is to develop **NearDrop**, a high-performance, cross-platform file-sharing application built with **Flutter**. NearDrop is designed to transcend the capabilities of contemporary tools like Xender by offering a "Link to Windows" style integration between mobile and desktop environments. The application must provide a seamless experience across Android, iOS, Windows, macOS, and Linux, functioning effectively in both offline and online scenarios. The ultimate goal is to achieve a five-star user rating on the Google Play Store through a combination of technical excellence and a user-centric interface.

Core Functional Specifications

Feature Category	Description	Technical Implementation
Hybrid Connectivity	Enables file transfers regardless of internet availability.	Wi-Fi Direct, BLE, mDNS, and WebRTC Relay.
NearLink Integration	Provides deep synergy between mobile devices and personal computers.	QR Pairing, Cross-Device Clipboard, and Notification Sync.
5-Star UI/UX	Delivers a modern, visually stunning, and intuitive interface.	Bento-box Layout, Glassmorphism, and Rive Animations.
Security & Privacy	Ensures all user data remains private and secure during transit.	AES-256 End-to-End Encryption and Granular Permissions.

Connectivity and Transfer Logic

NearDrop utilizes a **Smart Switch** connectivity layer that dynamically selects the most efficient transfer protocol. In local environments, the app prioritizes **Wi-Fi Direct** and **mDNS** for high-speed peer-to-peer (P2P) transfers. When devices are on separate networks,

the system transitions to an **Online Relay** powered by **WebRTC Data Channels**. This dual-path architecture ensures that users can share files of any size without being tethered to a specific network configuration.

The NearLink Ecosystem

The **NearLink** feature is the cornerstone of the application's unique value proposition. It facilitates a persistent connection between a user's smartphone and their PC. This integration includes a **Real-Time Clipboard Synchronizer**, allowing users to copy text on one device and paste it on another instantly. Furthermore, the **Notification Mirroring** system enables users to view and respond to mobile alerts directly from their desktop, minimizing digital distraction and enhancing productivity.

Technical Architecture and Requirements

The application shall follow a clean, modular architecture to ensure maintainability and scalability. **Riverpod** is the designated state management solution, providing a robust and testable framework for complex application states. For data persistence, **Isar** or **Drift** will be employed to handle lightning-fast indexing of the device's file system and transfer history.

Component	Recommended Technology
State Management	Riverpod / BLoC
Local Database	Isar (NoSQL) or Drift (SQL)
Networking	Dio (HTTP) & Flutter WebRTC (P2P)
Animations	Rive & Lottie
UI Framework	Flutter 3.x with Adaptive Design

Implementation Strategy for Claude Code

To build NearDrop using **Claude Code**, the development process should be executed in logical phases to ensure structural integrity and feature parity across platforms.

- Foundational Setup:** Initialize a multi-platform Flutter project and establish the directory structure, focusing on a feature-first approach (e.g., `features/transfer`, `features/link`, `core/network`).

2. **Connectivity Layer Development:** Implement the discovery services using mDNS and Bluetooth, followed by the socket-based transfer logic for both local and remote channels.
 3. **NearLink Service Implementation:** Develop the background services for clipboard monitoring and notification interception, ensuring proper permission handling on both Android and Desktop.
 4. **UI/UX Orchestration:** Construct the "Bento" dashboard using a responsive grid system. Integrate **Rive** animations to provide visual feedback during device discovery and file transfer processes.
 5. **Optimization and Security:** Apply AES-256 encryption to the data stream and optimize the app's performance to minimize battery consumption and maximize transfer speeds.
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Branding and Identity

The brand identity of **NearDrop** is defined by its reliability, speed, and futuristic aesthetic. The logo features a stylized drop shape intertwined with wireless signal waves, symbolizing the fluid and wireless nature of modern data sharing. The color palette consists of **Electric Blue** and **Deep Purple**, evoking a sense of high-tech sophistication.

Instruction for Claude Code: "Act as a Lead Flutter Architect. Using this blueprint, generate the complete, production-ready codebase for NearDrop. Prioritize modularity, security, and a 5-star user experience. Begin with the core architecture and connectivity services before moving to the UI implementation."